



Mathematical modeling of in situ synthesis of metal sulfides by sulfate-reducing bacteria and treatment of acidic mine drainage

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ABSTRACT

Acidic mine drainage (AMD) poses a significant environmental threat due to its high acidity and heavy metal (HM) content. In this study, sulfate-reducing bacteria (SRB) were employed to remediate AMD through a multi-cycle treatment strategy. SRB not only reduced sulfate to sulfide, enabling the precipitation of Fe^{3+} and Zn^{2+} as insoluble sulfides, but also facilitated the in situ biosynthesis of metal sulfides. The proposed treatment process led to substantial improvements in AMD water quality, with pH increasing from 5.5 to 8.0, Eh decreasing from +86.8 to −200 mV, an almost 50 % reduction in total organic carbon (TOC), and sustained sulfate depletion. Notably, complete removal (100 %) of Fe^{3+} and Zn^{2+} was achieved. Microbial community analysis revealed a compositional shift toward SRB-dominated populations after sludge domestication. To support and optimize this bioremediation process, an ordinary differential equation (ODE) model was developed to simulate the dynamics of SRB growth, sulfate reduction, and metal sequestration. Stability analysis confirmed the existence of a positive equilibrium point, where numerical simulations aligned with experimental observations. Sensitivity analysis further identified microbial growth rate and sulfate reduction efficiency as the dominant parameters most influencing system performance. Overall, this work not only demonstrated the dual benefit of SRB in metal sulfides nanoparticle synthesis and wastewater detoxification, but also established a predictive modeling framework to enhance the long-term applicability and design of SRB-based AMD treatment systems in the future.

1. Introduction

Acidic mine drainage (AMD) results from the oxidative dissolution of metal sulfides when exposed to water and oxygen, a process often exacerbated by human mining activities [1]. This phenomenon not only leads to the acidification of water bodies but also causes the release of toxic heavy metals (HMs) such as iron, zinc, and copper, which pose significant risks to surrounding ecosystems and human health [2]. Moreover, the high toxicity of metals in AMD exacerbates environmental degradation and complicates efforts to restore contaminated sites [3]. Recent studies have shown that indigenous microorganisms can accelerate the oxidative dissolution of metal sulfides, further intensifying AMD's harmful effects on the environment [2,4]. In contrast, sulfate-reducing bacteria (SRB) present a promising bioremediation strategy for AMD. SRB are capable of reducing sulfate to sulfide,

which subsequently reacts with HMs to form insoluble metal sulfides, effectively removing these toxic metals from wastewater [5,6]. Over the past few decades, culture-dependent methods have revealed the roles of AMD microorganisms in metal sulfide oxidation and acid production. More recently, culture-independent approaches have provided deeper insights into the microbial diversity and metabolic potential of AMD communities, advancing the understanding of their functions and interactions in AMD ecosystems [7,8]. The ability of SRB to mitigate the negative impacts of AMD by both neutralizing acidity and precipitating HMs makes them a key player in sustainable water treatment strategies.

Recently, metal sulfides such as ferrous sulfide nanoparticles have gained increasing attention in the field of environmental remediation due to their high surface area, strong reactivity, and affinity for HMs [9,10]. Compared with chemically synthesized counterparts, SRB biosynthesized metal sulfides exhibit better dispersibility and stability

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under natural conditions, because they are formed in biologically controlled redox environments [4]. These biogenic metal sulfides serve not only as metal scavengers via co-precipitation and adsorption, but also as reducing agents which participate in contaminant transformation [5,11]. However, the mechanisms and efficiency of in situ metal sulfide formation by SRB under AMD conditions remains underexplored, and scant studies have integrated such biological processes with modeling approaches to provide new opportunities for understanding and optimizing such nanoscale transformations during AMD treatment.

Wastewater treatment plants are indispensable components of modern urban infrastructure, playing a pivotal role in safeguarding and enhancing the water environment [12]. They are responsible for the purification of water that has been contaminated by various pollutants, ensuring that it meets environmental standards before being discharged back into natural water bodies. However, the process of optimizing the design and operational management of these facilities is fraught with challenges. It is essential to strike a balance between maintaining high water quality standards and minimizing operational expenditures, including reducing resource wastage and energy consumption [13]. Achieving this delicate equilibrium is crucial for the sustainability and cost-effectiveness of wastewater treatment operations. One of the most promising approaches to overcoming these challenges is the application of mathematical models for the simulation and control of wastewater treatment processes [14]. These models offer a powerful tool for gaining insights into the complex interactions occurring within the treatment system. They can provide a detailed and quantitative description of the myriad biological reactions taking place, enabling operators to predict the quality of the effluent produced by the plant [12]. Furthermore, mathematical models are invaluable for analyzing the operational status of the treatment plant, facilitating informed decision-making regarding process selection, management strategies, and the optimization of both plant design and operational efficiency, particularly in terms of energy conservation [15].

The development and application of mathematical models for microbial activated sludge systems, which are at the heart of many biological wastewater treatment processes, holds immense value for research into the improved design of water treatment systems [16,17]. The progression from traditional static models to more sophisticated dynamic models signifies a deepening understanding of the intricacies of wastewater treatment. Early static models, such as those developed by Eckenfelder, McKinnie, and Lawrence-McCarty, laid the groundwork for understanding the basic principles of activated sludge systems [18]. These were followed by the development of dynamic models, including the influential works of Andrews, the Water Research Centre (WRC), and the Activated Sludge Model (ASM) series, which have furthered the comprehension of the dynamic nature of these systems [18–20]. These models are not only crucial for enhancing the understanding of the fundamental principles and mechanisms underlying wastewater treatment but also serve as indispensable tools for practical applications. They allow researchers and operators to analyze various parameters based on empirical experimental data, enabling a more precise prediction of the behavior of different components within the aeration tank over time. This predictive capability is particularly valuable for adjusting process parameters in real-time, ensuring optimal treatment efficiency and compliance with water quality standards [18,21]. By providing a scientific basis for these adjustments, mathematical models contribute significantly to the advancement of wastewater treatment technology and the pursuit of more sustainable and efficient treatment processes.

Many previous studies [22,23] have shown that the application of SRB was beneficial in AMD treatment, enhancing both HM removal and neutralizing acidity. For instance, Guo et al. [23] investigated SRB-assisted remediation in the presence of zero-valent iron, while Zhang et al. [22] reviewed the various metabolic pathways influencing sulfate reduction. However, most of the existing studies to date have focused primarily on empirical observations, and the incorporation of dynamic

modeling to predict treatment performance or system behavior over time has not been conducted. The novelty of this study lies in the integration of experimental results with a robust ordinary differential equation (ODE) model based mathematical model, which not only captures SRB dynamics but also provides predictive insights for long-term system stability, parameter optimization, and sustainability. This unique modeling approach has been largely ignored in the SRB-related AMD remediation literature, where application here marks a key innovation of the work.

Despite existing research on mathematical models for wastewater treatment, challenges remain in enhancing the accuracy and applicability of these models and in integrating them with practical processes [17]. The main innovation of this study lies in the establishment of a mathematical model based on experimental procedures and data results specifically for SRB-mediated bioremediation of AMD. The model captures the interactions between SRB population growth, HM sequestration, and changes in environmental conditions. Through rigorous mathematical analysis, this study aims to determine the existence and stability of equilibrium points in the model, thereby providing insights into the robustness and sustainability of the treatment system. In addition, the feasibility of the model and the accuracy of the experimental results were validated through numerical simulations. The three main questions to be addressed in this study were: (1) How to improve the removal efficiency of HMs present in AMD? (2) How to ensure the stability and sustainability of SRB during AMD treatment? and, (3) How to optimize the process parameters used in the SRB treatment of AMD to improve treatment efficiency?

To address these issues, continuous treatment experiments were first conducted to investigate the role of SRB over multiple AMD treatment cycles. Techniques were employed, collectively with various parameter tests, to observe the dynamic changes in HM removal by SRB. Thereafter, a mathematical model was constructed to simulate the interactions between SRB population growth, HM sequestration, and any changes in environmental conditions. This model was then used to optimize the treatment process and improve HM removal efficiency. Finally, the stability of the system was analyzed through the mathematical model, and the existence and stability of equilibrium points were determined, providing theoretical support for the long-term application of SRB. This study not only contributes to a comprehensive understanding of the role of SRB in AMD bioremediation, but also provides a practical approach for the sustainable management of HM-contaminated wastewater.

2. Experimental section

The overall experimental methodology was specifically designed to link experimental observations with mathematical modeling. Laboratory-scale AMD treatment was conducted over multiple cycles, with key chemical and biological indicators monitored to evaluate SRB performance. These data were then used to parameterize and validate an ODE-based model that simulates SRB dynamics, sulfate reduction, and metal precipitation. Statistical and comparative analyses were performed to ensure the reliability of results.

2.1. Experimental setup

The reaction was studied in an up-flow anaerobic reactor (Fig. 1). The bottom one fifth of the reactor was loaded with AMD sediment to serve as a carrier for microbial growth. An effluent outlet was positioned on the top left side of the reactor, and a water tank was placed next to the reactor. Water was pumped into the reactor using a peristaltic pump. The entire reactor was placed on a constant temperature electromagnetic stirrer, which was periodically activated to ensure that the substrate within the reactor was evenly dispersed, and an intelligent temperature control electric heating belt was used to maintain internal reactor temperature at 30 °C, to ensure a suitable environment for SRB

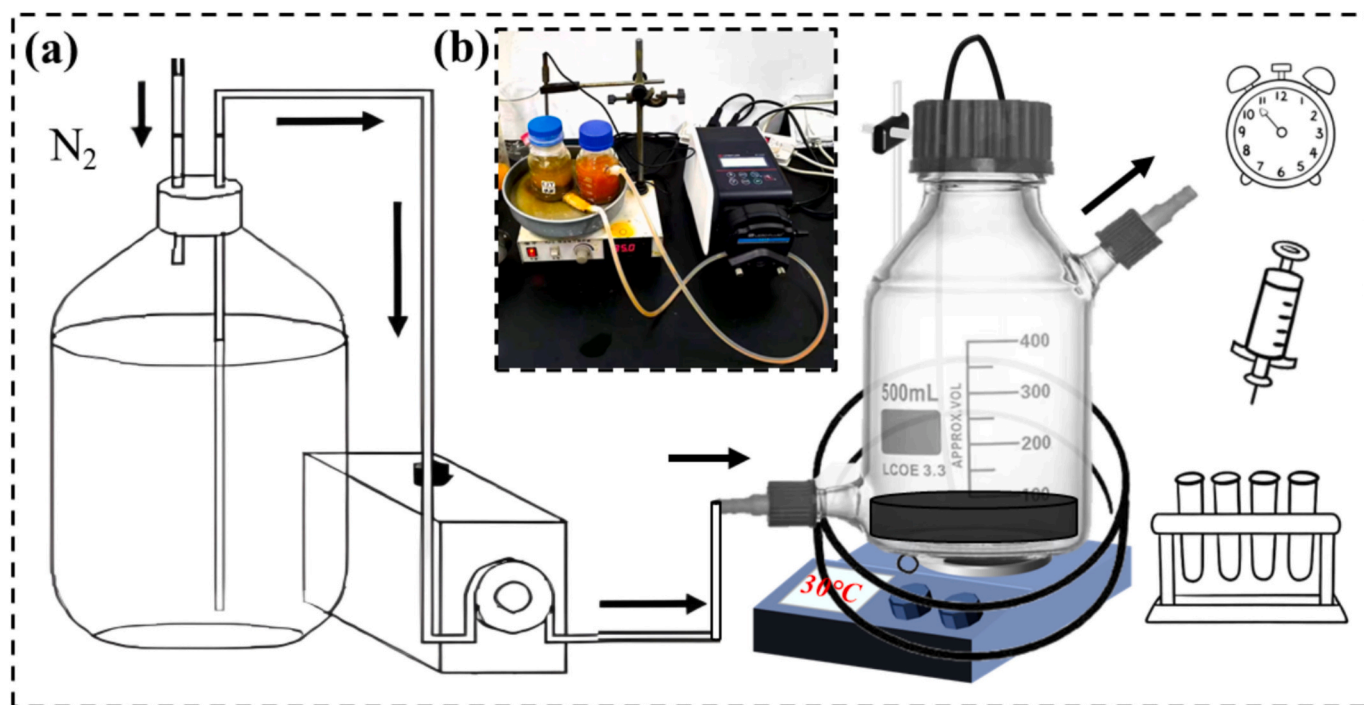


Fig. 1. Schematic of the up-flow anaerobic reactor (a) and an actual picture of the unit reaction (b).

proliferation.

2.2. Domestication of sludge and AMD background values

The process began with enrichment and domestication of SRB using an anaerobic granular sludge collected from the bottom of AMD at a sewage treatment plant in Longyan, Fujian Province. The initial volume of the inoculated sludge was one-fifth of the effective volume of the reactor. The domestication medium was prepared using sodium lactate as the carbon source, ammonium sulfate as the nitrogen source, and potassium dihydrogen phosphate as the phosphorus source, at a constant ratio of COD:N:P = 200:5:1. Sodium lactate was chosen as the primary carbon source because it has a well-established role as an efficient electron donor for SRB metabolism under anaerobic conditions. Ammonium chloride and potassium phosphate were simply included to ensure sufficient nitrogen and phosphorus availability for microbial growth. Magnesium and calcium salts contribute to enzymatic cofactor availability and osmotic balance, and the pH was adjusted to 5.5 to simulate mildly acidic conditions conducive to SRB activation [24]. Sulfate was present in the system as a complex mixture of ammonium sulfate, sodium sulfate, ferrous sulfate heptahydrate, and magnesium sulfate. Thereafter, trace elements were added (Table 1), and the pH adjusted to 5.5 with sodium carbonate. Temperature was also maintained at 30 °C during the domestication period. After one week domestication, a mixed culture of SRB capable of removing HM ions from AMD was obtained.

The initial selected background levels of HMs in AMD and other

physicochemical parameters before bioremediation are summarized in Table 2, thus allowing for accurate assessment of the efficiency of SRB in removing these contaminants.

2.3. Characterization and methods of analysis

Changes in the morphology of the sulfate-reducing active sludge before and after domestication were observed using a scanning electron microscope (SEM Regulus 8100 model, manufactured by HITACHI). Additionally, a third-generation sequencing platform was utilized, targeting the variable regions within the 16S rRNA gene's transcribed spacers, where variations in the type and quantities of the sequences, was used to monitor changes in the types and abundance of microbial communities in the sludge.

During the treatment of AMD with activated sludge, reactor water samples were collected daily to measure various parameters. Changes in pH and Eh (redox potential) were measured using a pH meter (Shanghai Yi Dian Scientific Instrument Co., Ltd., PHS-3E) fitted with an appropriate electrode (a pH electrode or an Eh electrode). Total organic carbon (TOC) was measured using a total organic carbon analyzer (Analytical Instruments Corporation of China, TOC-L CPH). Sulfate ions were measured using ion chromatography (Thermo Fisher Scientific, USA, AQ-1100 model). Iron and zinc ions were determined using an inductively coupled plasma optical emission spectrometer (ICP-OES, PerkinElmer, USA, Optima 8000).

All experimental measurements, including pH, Eh, TOC, SO_4^{2-} ,

Table 1
Media formulations of SRB.

Medicaments	Dose (mg/L)	Medicaments	Dose (mg/L)
$C_3H_5NaO_3$	3000	$FeSO_4 \cdot 7H_2O$	2000
$(NH_4)_2SO_4$	75.00	$CuSO_4$	29.10
KH_2PO_4	15.00	$CoCl_2 \cdot 6H_2O$	0.200
Na_2SO_4	2924	$ZnSO_4 \cdot 7H_2O$	44.30
$MgSO_4$	25.00	H_3BO_3	1.000
$CaCl_2$	30.00	$MnSO_4$	20.00

Table 2
Selected initial physiochemical AMD parameters.

Items	Values (mg/L)
pH	5.5
SO_4^{2-}	57.05
TOC	811.2
Eh	86.75
Fe^{3+}	627.04
Zn^{2+}	11.363

$\text{Fe}^{3+}/\text{Fe}^{2+}$, and Zn^{2+} concentrations, were conducted in triplicate ($n = 3$) and data presented as mean $\pm$ standard deviation (SD). Statistical significance of changes across treatment cycles were assessed using Origin software [25].

2.4. Mathematical model formulation

A mathematical model was constructed to gain a better understanding of the dynamics of HM removal by SRB during wastewater treatment, where the interactions between microorganisms (X), sulfate (S), sulfide (P), and HM (M) were described. The model's ordinary differential equations are shown in the following system (1). The model captures interactions between SRB population dynamics, sulfate reduction, sulfide production, and HM sequestration, where similar modeling frameworks have been adopted in previous studies on anaerobic bioreactors and metal remediation processes [22].

$$\begin{cases} \frac{dX}{dt} = \frac{QX_0}{V} + rX\left(1 - \frac{X}{K}\right) - k_d X \\ \frac{dS}{dt} = \frac{QS_0}{V} - k_x SX - \frac{QS}{V} \\ \frac{dP}{dt} = k_s SX - k_p PM \\ \frac{dM}{dt} = \frac{QM_0}{V} - k_m PM - \frac{QM}{V} \end{cases} \quad (1)$$

where X (cells/mL), S , P , and M (mg/L) represent the concentrations of microorganisms, sulfate, and sulfide, a HM, respectively, at time t (d). Q (L/day) is the flow rate of wastewater; V (L) is the wastewater volume in the reactor; X_0 and S_0 represent the influent concentrations of microorganisms (cells/mL) and sulfate (mg/L), respectively; r (1/day) is the growth rate of microorganisms; K is the maximum concentration of microorganisms; k_d (1/day) is the decay constant of microorganisms; X_e (cells/mL) is the effluent concentration of microorganisms; k_x (dimensionless) is the rate constant for sulfate consumption by microorganisms; k_s (L/mg-day) is the rate constant for the biological conversion of sulfate (S) to sulfide (P) by microorganisms (X); k_p and k_m (dimensionless) are rate constants representing the reactions between sulfide and HM,

including reduction and precipitation into insoluble metal sulfides [4]. These parameters were selected based on conventional assumptions in activated sludge and sulfate-reducing systems [18], ensuring both biological relevance and computational feasibility.

To solve the ODE system, initial conditions were assigned to each state variable based on experimental measurements. Specifically, the initial SRB biomass concentration $X(0)$, sulfate concentration $S(0)$, sulfide concentration $H(0)$, and HM ion concentration $M(0)$ were set to match the values at the start of the treatment cycle. These include: $X(0) = 1.0 \times 10^7$ cells/mL (estimated from inoculum density); $S(0) = 57$ mg/L (initial SO_4^{2-} concentration from AMD input); $P(0) = 0$ mg/L (no initial sulfide) and $M(0) = 627$ mg/L for Fe^{3+} and 9.4 mg/L for Zn^{2+} . These initial values provided the boundary framework for numerical simulation in Matrix Laboratory (MATLAB), and were kept constant for all baseline simulations unless varied during sensitivity analysis.

3. Results and discussion

3.1. Changes in activated sludge before and after domestication

Morphological changes in the activated sludge before (Fig. 2a) and after (Fig. 2b) domestication show that before domestication, the surface of the sludge was predominantly composed of coarse particles and large organic molecules, with relatively few bacterial cells. However, after domestication the surface of the sulfate-reducing active sludge became populated with a significant number of SRB bacteria due to the enrichment effect of the SRB culture medium [26]. The increase in the number of observed bacteria the sludge surface, which were mainly spherical and rod-shaped, indicated successful SRB community enrichment and growth.

There was a significant change in the composition of the bacterial community abundance in the reactor activated sludge before and after domestication (Fig. 2c). Before domestication, the bacterial community abundance in the reactor was primarily dominated by only two species, *Acidibacillus* and *Acidithiobacillus*, with *Acidibacillus* being more prevalent than *Acidithiobacillus*. After domestication, the activated sludge became enriched with the three main SRB bacterial species, *Thermincola* [27], *Desulfosporosinus* [28], and *Clostridium sensu stricto* 12 [29]. All of these three species were significantly enriched, with their relative

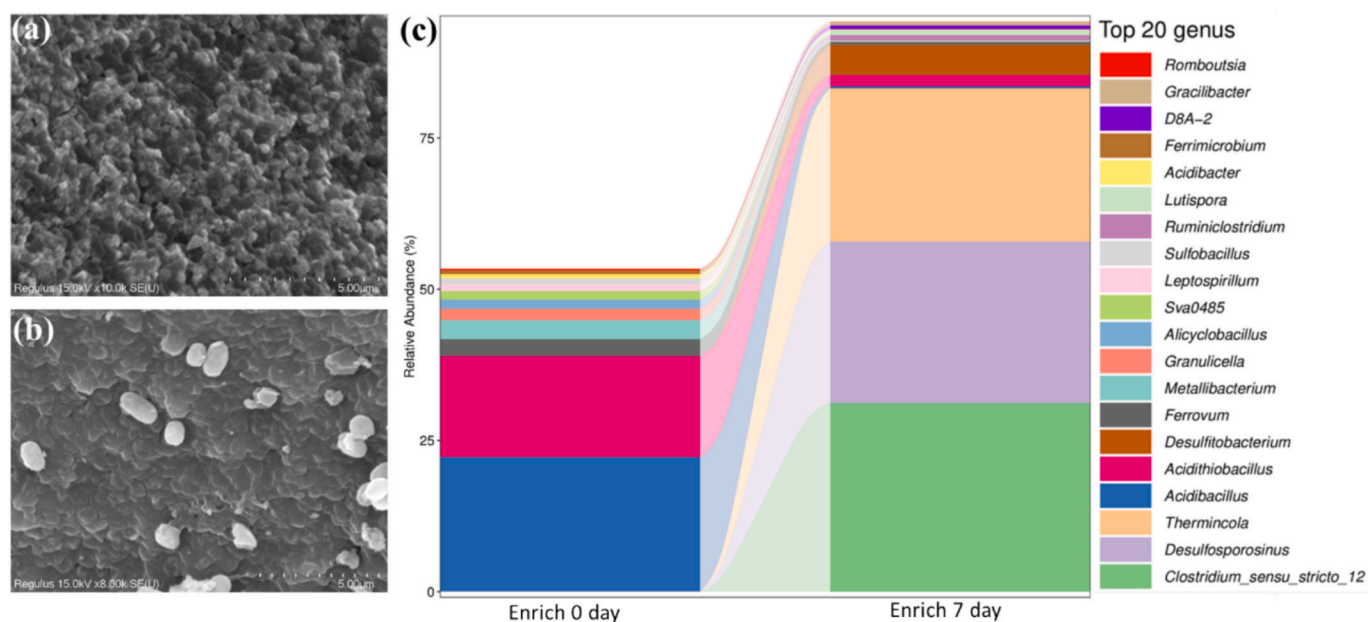


Fig. 2. Morphological and microbial shifts before and after sludge domestication. (a) SEM of raw sludge; (b) SEM after SRB enrichment; (c) relative abundance of microbial genera pre- and post-domestication.

proportions becoming approximately equal, at a ratio of about 1:1:1, making them the dominant post-reaction species. Ultimately, after a series of reactions, the bacterial biomass after the reaction was roughly twice that of the biomass before the reaction, indicating an enhancement in the microbial population and activity within the sludge.

The observed enrichment of the three SRB species can mechanistically be explained by their metabolic adaptability under anaerobic and sulfate-rich conditions [28]. These bacteria all possess the dissimilatory sulfate reduction pathway, where sulfate (SO_4^{2-}) is first activated to adenosine 5'-phosphosulfate (APS) by sulfate adenylyltransferase, then reduced to sulfite by APS reductase, and finally converted to sulfide (S^{2-}) via dissimilatory sulfite reductase [30,31]. This metabolic process requires electron donors such as lactate or other organic carbon sources, which were amply supplied in the domestication medium. The anaerobic up-flow reactor also maintained reducing conditions (as evidenced by decreasing Eh), creating an ideal environment for SRB proliferation. Furthermore, the competitive exclusion of acidophilic bacteria like *Acidibacillus* may be attributed to the gradual pH elevation caused by SRB activity, favoring neutrophilic or alkaliphilic SRB. Together, these factors synergistically promoted the dominance of SRB over other microbial groups during domestication.

3.2. Factors of change in the treatment process

During the treatment of AMD by SRB the detailed tests on key parameters (Sections 3.2.1 to 3.2.4), including SO_4^{2-} , TOC, pH, and Eh, allowed a deeper understanding of the mechanisms by which SRB facilitate bioremediation in AMD. Specifically, this allowed for 1) an assessment of the effectiveness and stability of the treatment process, and 2) provided accurate source data to support the development and optimization of a mathematical model. More fundamentally, by

monitoring these parameters, the mechanisms by which SRB reduced the acidity and toxicity of AMD through metabolic processes while removing HM could be elucidated. These parameters also confirm that SRB maintain stable biological activity over multiple treatment cycles, thereby laying the foundation for the long-term effective management of AMD.

3.2.1. SO_4^{2-}

Sulfate ion concentrations steadily decreased with the increasing number of treatment cycles (Fig. 3a). While the initial rapid decrease suggested active microbial sulfate reduction, later stages demonstrated a much slower but consistent decline. This may be because the initial AMD lacked sufficient organic matter to support SRB growth. As AMD was periodically introduced into the reactor, the accumulated organic matter gradually became adequate for SRB metabolism, leading to increasingly vigorous SRB activity and an accelerated rate of sulfate reduction. This pattern confirms SRB adaptation and sustained metabolic activity [5,6]. The significant fluctuations within the broad overall decline in sulfate concentrations suggest a dynamic interaction between microbial activity and chemical conditions. The periodic drops likely represent an intensified microbial reduction of sulfate to sulfide, while the rises could be due to temporary substrate limitations or competition from other microorganisms. The overall trend signifies a progressive and effective reduction of sulfate ions, with later stages showing a more consistent decrease. This suggested the SRB community was stabilizing, leading to a more reliable rate of sulfate reduction. In essence, (Fig. 3a not only confirms SRB's role in mitigating AMD but also emphasizes the necessity for precise environmental control to optimize their bio-remedial efficacy. This decline over time is consistent with SRB participating in a dissimilatory sulfate reduction pathway, where sulfate serves as a terminal electron acceptor, producing sulfide under anaerobic conditions

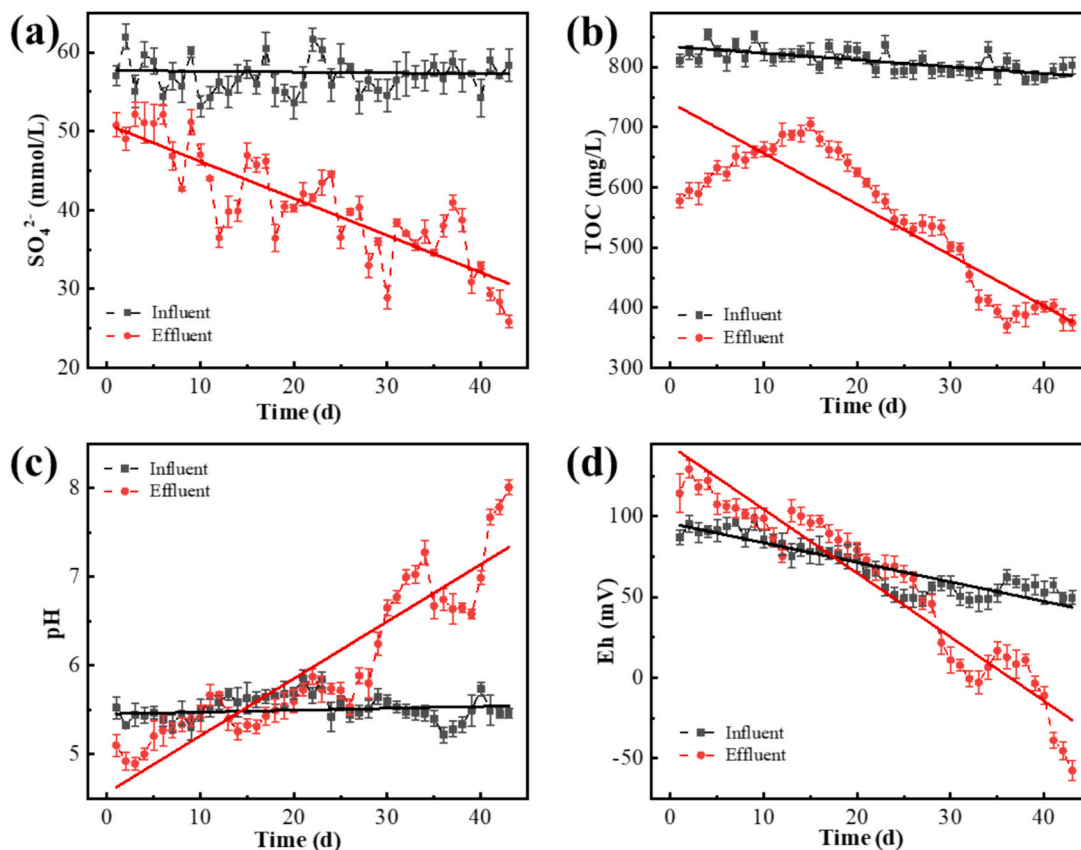


Fig. 3. Changes in key water quality parameters during SRB treatment of AMD. (a) Decline in SO_4^{2-} ; (b) reduction in TOC; (c) increase in pH; (d) decrease in Eh. Data represent mean $\pm$ SD ($n = 3$); solid lines represent linear regression fits using Origin.

[22]. The gradual stabilization over time suggests a balance between substrate availability and microbial activity.

3.2.2. TOC

During domestication total organic carbon (TOC) concentrations remained relatively constant, fluctuating around 800 mg/L (Fig. 3b). However, under the influence of SRB, a series of continuous reactions resulted in the effluent TOC concentrations being consistently lower than that of the influent concentrations. This indicates that SRB was directly involved in reducing the concentration of TOC in the wastewater [32]. This may be attributed to the metabolic activity of SRB utilizing organic carbon present in the AMD. Over successive treatment cycles, the gradual accumulation of residual organics enabled the establishment of an internal carbon cycle within the reactor, thereby sustaining SRB activity without the need for external carbon supplementation [1,25]. As the reaction progresses, particularly in the later stages under optimal conditions, the effluent TOC concentration was reduced to approximately half of the influent concentrations. This significant decrease demonstrated substantial consumption of TOC in AMD by SRB. In turn, the organic matter present in AMD contributed to the growth of SRB, highlighting a synergistic relationship where the bacteria find sustenance in the AMD's organic content while simultaneously mitigating its environmental impact. The decrease in TOC was attributed to increased SRB metabolism, in which organic carbon (e.g., lactate) is oxidized to supply electrons for sulfate reduction [32]. This closed-loop use of organic carbon reduces reliance on external carbon inputs, thereby improving the sustainability of the system.

3.2.3. pH

Changes in pH occurred during AMD treatment (Fig. 3c). Initially, the influent pH remained relatively stable around 5.5, indicating a mildly acidic environment. As treatment progressed, the average pH gradually increased, reflecting a transition toward alkaline conditions. This upward trend suggests that SRB effectively neutralized acidity through their metabolic activity [24]. SRB metabolic processes consume sulfate ions and convert them into sulfide, which can then react with metals to form insoluble metal sulfides, thereby lowering the overall acidity [4,5]. In the latter stages of the reaction, effluent pH increases to pH 8 and becomes significantly higher than that of the influent pH. This significant increase to near-neutral or slightly alkaline pH indicates that SRB plays a substantial role in mitigating AMD acidity [3]. The pH changes observed during AMD treatment underscore the beneficial impact of SRB in reducing acidity and promoting more balanced environmental conditions. The rise in pH corresponds to the consumption of protons during sulfate reduction and the formation of metal sulfide precipitates, which reduce the free acidity of the solution [4]. Similar pH elevation trends have been reported in SRB-based AMD treatments [3].

3.2.4. Eh

Significant temporal changes in the redox potential (Eh) occurred during AMD treatment (Fig. 3d). Initially, the Eh of the effluent samples was higher than that of the influent samples, particularly in the early and middle stages of the reaction, indicating a relatively more oxidizing environment. However, in the later stages of the reaction, the Eh of the effluent solution decreased well below that of the influent solution, even becoming negative, signifying a shift from acidic to weakly alkaline conditions due SRB activity. This transition suggests that SRB was successfully altering the chemical environment, by reducing the water's acidity [24]. Concurrently, the overall decline in Eh throughout the treatment process indicates a progressive reduction in the system's oxidizing capacity, further supporting the activity of SRB in lowering the redox potential of the AMD [33]. This shift toward more reducing conditions is a positive indicator of ongoing biological sulfate reduction, which plays a critical role in detoxifying and remediating AMD. The observed decrease in Eh reflects the establishment of a strongly reducing, anaerobic environment—an essential prerequisite for effective

SRB-mediated sulfate reduction [33].

3.3. Changes in heavy metal speciation

During the domestication process, the concentrations of $\text{Fe}^{2+}/\text{Fe}^{3+}$ (Fig. 4a) and Zn^{2+} (Fig. 4b) declined significantly due to SRB-mediated adsorption and precipitation. The removal efficiencies of Fe^{3+} and Zn^{2+} increased steadily with treatment cycles, reaching 100 % within 28 days and 13 days, respectively, indicating successful SRB adaptation and high metabolic activity. In contrast, control groups without SRB inoculation exhibited negligible metal removal (<5 %), confirming that the observed performance was primarily driven by biological sulfate reduction and subsequent metal sulfide precipitation [4]. These results not only demonstrate the effectiveness of microbial domestication but also highlight the system's potential for long-term stable operation. The sustained removal capacity across cycles suggests that SRB maintain consistent adsorption and reduction capabilities, making them a reliable tool for heavy metal remediation in industrial effluents [5,23].

3.4. Analysis of the model

While the experimental results discussed in Sections 3.1 to 3.3 reveal some key dynamic changes in the treatment of AMD by SRB, several limitations still exist. Firstly, these experimental data demonstrate only the relatively short-term effects of SRB on HM removal and environmental parameter regulation, but lack a deeper understanding of long-term stability and system behavior. Secondly, the complexity and variability of experimental conditions make it difficult to accurately predict treatment outcomes and system responses under different operational conditions. Additionally, the interrelationships between the observed phenomena and the results were not entirely clear, requiring a more systematic analysis to uncover underlying mechanisms. To address these issues, the application of a mathematical model is crucial. By constructing a model, the complex interactions between SRB population growth, HM sequestration, and environmental condition changes can be simulated, allowing for predictions of system behavior under various conditions and providing a theoretical basis for optimizing treatment processes and improving efficiency.

3.4.1. Existence of an equilibrium point

Considering the specific initial case where the concentration of microorganisms at the time of water intake is very low and negligible (i.e., let $X_0 = 0$) allows system (1) to be simplified as follows:

$$\begin{cases} \frac{dX}{dt} = rX \left(1 - \frac{X}{K} \right) - k_d X \\ \frac{dS}{dt} = \frac{QS_0}{V} - k_x S X - \frac{QS}{V} \\ \frac{dP}{dt} = k_s S X - k_p P M \\ \frac{dM}{dt} = \frac{QM_0}{V} - k_m P M - \frac{QM}{V} \end{cases} \quad (2)$$

To effectively study the dynamical properties of system (2), requires determination of the system's equilibrium point [34]. In this system, the equilibrium point represents the system's steady state under specific conditions. In the specific case of SRB applied for AMD bioremediation, the presence of an equilibrium point means that there is a combination of environmental conditions and bacterial activity that allow the system to reach a relatively stable state. Calculations conclude that the positive equilibrium points in system (2) are as follows (3)–(6):

$$X^* = \frac{K(r - k_d)}{r} \quad (3)$$

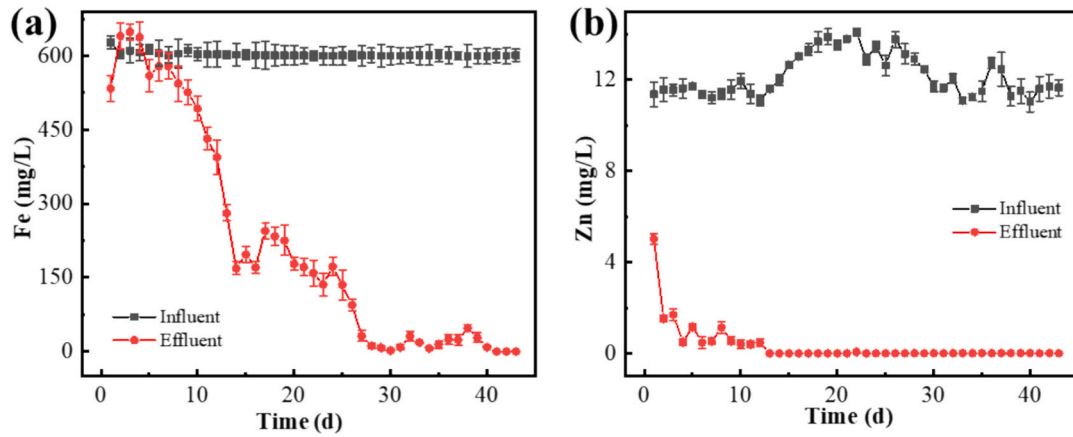


Fig. 4. Temporal variation in the removal efficiency of Fe^{3+} (a) and Zn^{2+} (b) during repeated SRB treatment cycles. All values represent mean $\pm$ SD ($n = 3$).

$$S^* = \frac{QS_0}{V\left(k_x X^* + \frac{Q}{V}\right)} \quad (4)$$

$$M^* = M_0 - \frac{k_s k_m S^* X^* V}{Q k_p} \quad (5)$$

$$P^* = \frac{k_s S^* X^*}{M^* k_p} \quad (6)$$

Clearly, according to the theory of stability of differential equations [35], system (2) possesses a positive equilibrium point when condition (7) is satisfied:

$$\left(\lambda - r + \frac{2r}{k} X^* + k_d\right) \left(\lambda + k_x X^* + \frac{Q}{V}\right) \left[\lambda^2 - \left(-k_p M^* - k_m P^* - \frac{Q}{V}\right) \lambda + \left(k_p M^* \left(k_m P^* + \frac{Q}{V}\right) - k_p k_m P^* M^*\right)\right] = 0 \quad (9)$$

$$(H_1) : r > k_d, M_0 > \frac{k_s k_m S^* X^* V}{Q k_p} \quad (7)$$

From a practical application perspective, this condition means that 1) when the birth rate of the microorganisms exceeds the mortality rate and 2) the influent concentration of HMs exceeds the equilibrium concentration, the system will have a positive equilibrium point. The existence of a stable equilibrium point indicates that over time the SRB population, sulfate, and HM concentrations can all reach a predictable steady state. This is critical for reactor design, as it implies that after an adaptation period, the system can maintain high performance with minimal intervention. In practical applications, such stability supports the use of automated or semi-autonomous operations, particularly in remote or decentralized treatment settings.

3.4.2. Stability analysis of the positive equilibrium point

The system's equilibrium point indicates a possible stable condition for the system. To forecast how the system will behave in the future, it is necessary to analyze the stability of the equilibrium point [36,37]. Through computation, the Jacobi matrix for system (2) can be derived, as shown in Eq. (8):

$$J = \begin{bmatrix} r - \frac{2r}{k} X^* - k_d & 0 & 0 & 0 \\ -k_x S^* & -k_x X^* - \frac{Q}{V} & 0 & 0 \\ k_s S^* & k_s X^* & -k_p M^* & -k_p P^* \\ 0 & 0 & -k_m M^* & -k_m P^* - \frac{Q}{V} \end{bmatrix} \quad (8)$$

The corresponding characteristic equation of the Jacobi matrix for the system at the positive equilibrium point is shown in (9):

The eigenvalues are then (11)–(13):

$$\lambda_1 = r - \frac{2r}{k} X^* - k_d = -(r - k_d) < 0 \quad (10)$$

$$\lambda_2 = -k_x X^* - \frac{Q}{V} < 0 \quad (11)$$

$$\lambda_3 + \lambda_4 = -k_p M^* - k_m P^* - \frac{Q}{V} < 0 \quad (12)$$

$$\lambda_3 \lambda_4 = k_p M^* \frac{Q}{V} > 0 \quad (13)$$

According to the Routh-Hurwitz criterion [38], the positive equilibrium point of system (2) is locally asymptotically stable. That is, since the real part of all four eigenvalues is negative, the solution of the system will eventually converge to the positive equilibrium point. This stability has significant practical implications. In wastewater treatment processes, a stable system ensures the consistent and efficient performance of SRB. It allows for the effective removal of HMs and the modulation of environmental parameters, leading to improved water quality. For example, a stable system can maintain the appropriate HMs. Additionally, stability helps to minimize fluctuations in the treatment process, reducing the risk of unexpected failures and ensuring the reliability of the treatment system. This is particularly important in industrial applications where consistent and high-quality wastewater treatment is

Table 3

Parameter definitions and values used in the ODE-based model of sulfate-reducing bacteria (SRB) dynamics during AMD remediation.

Parameter	Symbol	Values or range	Units	Description
Microbial growth rate	r	2 (2–10)	1/day	Maximum specific growth rate of SRB
Carrying capacity	K	1.0×10^9	cells/mL	Max SRB concentration in reactor
Biomass decay rate	k_d	0.06	1/day	SRB endogenous respiration or death rate
Sulfate consumption rate	k_x	0.38	Dimensionless	The rate constant for the consumption of sulfate by microorganisms
Sulfate reduction rate constant	k_s	0.004	L/mg-day	Rate constant for sulfate-to-sulfide conversion
Reaction rate constant of metal sulfides	k_p	0.8	Dimensionless	The rate constant for the reaction between sulfide (P) and HM (M) that results in the reduction of sulfide
Reaction rate of metal sulfides	k_m	1.8	Dimensionless	The rate constant for the reaction between sulfide and HM to form insoluble products.
The duration	$\frac{V}{Q}$	4	day	The duration of the water coming out

Note: The numerical values of the parameter were determined either experimentally, or sourced from the literature, or adjusted during model calibration.

essential for environmental protection and regulation compliance.

3.4.3. Numerical simulation

To validate the accuracy of the model, microbial sulfate, sulfur ion, and HM concentrations were obtained from experimental analysis and used as input data. These parameters are summarized in Tables 2 and 3:

Thereafter, the MATLAB numerical software was used to solve the model. MATLAB is a widely used tool in scientific computing and has

been proven effective in solving various mathematical models in many research fields [39]. The model was integrated numerically using the ode45 function, because the ode45 function is a popular choice for solving ordinary differential equations numerically due to its efficiency and accuracy [40]. The dynamic curves of the concentration of each component over time were thus obtained (Fig. 5).

The simulation images (Fig. 5), showed that the individual concentrations corresponded to the actual values of the experiment when the equilibrium state was reached. This result is consistent with previous studies that demonstrated the reliability of numerical simulations in validating mathematical models [41]. Therefore, this also verifies the accuracy and reliability of the system. Specifically, a rapid decrease in SO_4^{2-} was observed at the beginning of the reaction, followed by stabilization, consistent with the reduction of sulfate by SRB. Meanwhile, the sulfide concentration gradually increased with the response, reflecting the biotransformation process of sulfate reduction to sulfide, which has also been documented in prior research [22,42]. The concentrations of HM ions (e.g., $\text{Fe}^{2+/3+}$ and Zn^{2+}) were also significantly reduced and eventually tended to zero, which is direct evidence of the high efficiency of SRB in removing HM ions.

The simulation results (Fig. 5) were compared with the experimental data (Figs. 3a, 4), revealing a consistent downward trend in both sulfate and heavy metal concentrations. Notably, the model also predicted complete removal of heavy metals over time, with concentrations approaching zero. In addition to reproducing experimental trends, the model offers the advantage of predicting unmeasured parameters. For example, it shows that the SRB population initially grows rapidly and then stabilizes as the system reaches metabolic equilibrium. Sulfate levels also tend to stabilize as SRB activity reaches a steady state, while sulfide concentrations gradually increase due to sulfate reduction. Overall, the model not only aligns well with experimental observations but also provides predictive insights into parameters not directly measured, thereby reinforcing the proposed mechanism of heavy metal removal.

3.4.4. Sensitivity analysis

To further understand the robustness and sensitivity of the proposed model, a sensitivity analysis was conducted by varying key parameters in the system. This analysis provides insights into how changes in

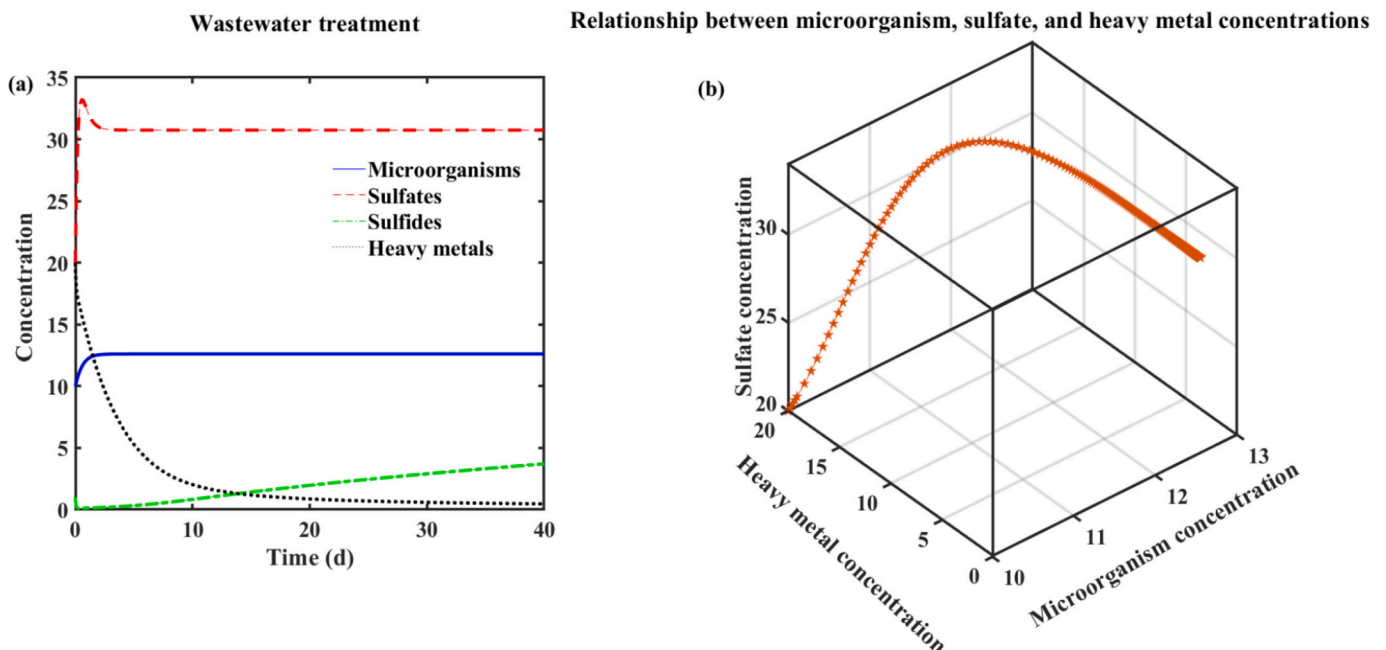


Fig. 5. Time sequence diagram (a) and phase diagram (b) of system (2).

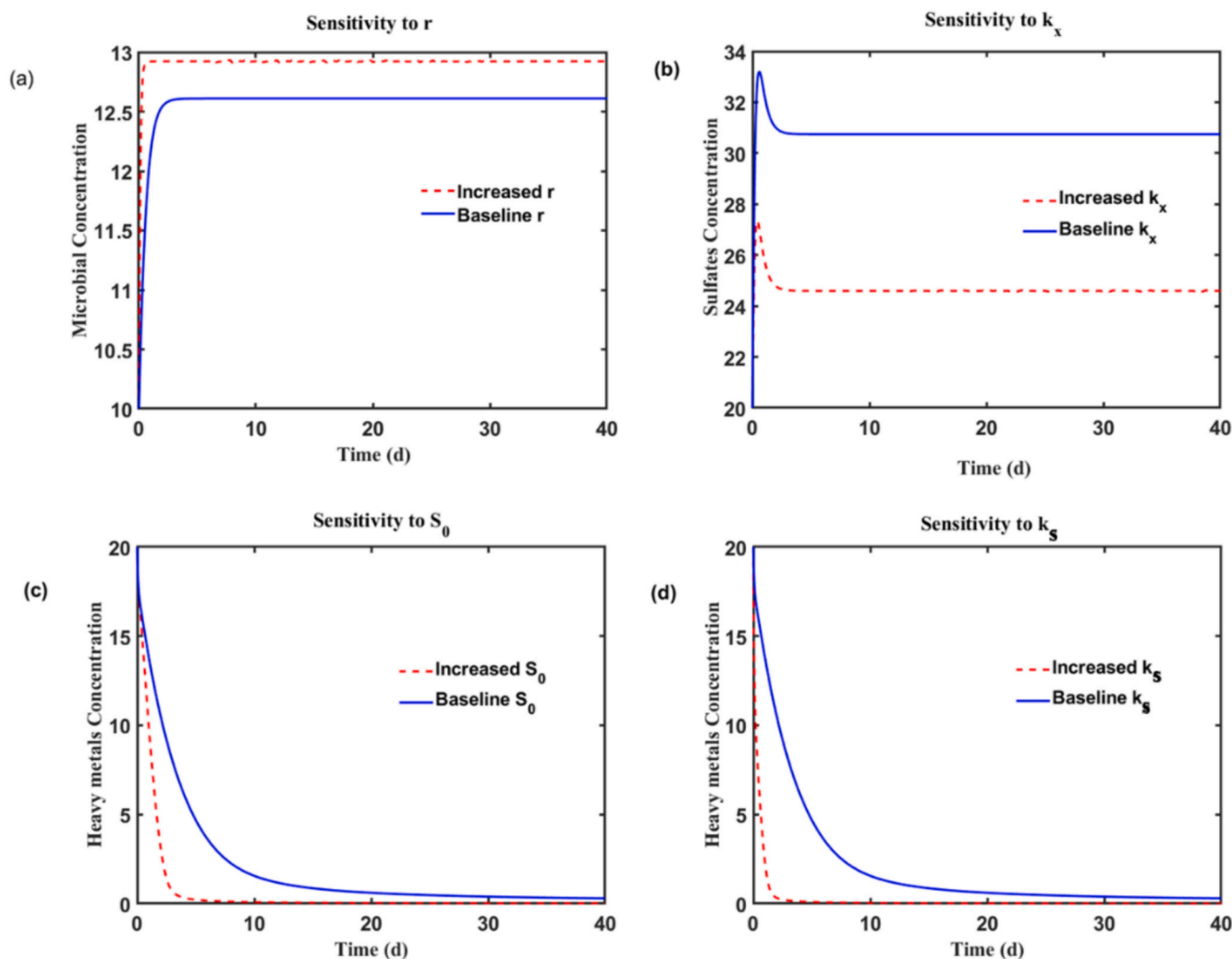


Fig. 6. Sensitivity analysis of system (2) with respect to parameters r (a), k_x (b), S_0 (c) and k_s (d).

environmental conditions or microbial activities can affect the dynamics of HM removal and the overall treatment process.

When the microbial growth rate (r) was varied from 2 to 10 while all other parameters were kept constant, the results indicated that an increase in the microbial growth rate leads to a faster reduction of sulfate and HMs (Fig. 6a). This result highlights the importance of maintaining optimal microbial growth conditions for enhanced bioremediation efficiency.

When the rate constant for sulfate reduction (k_x) was increased from 0.38 to 0.68, a faster decline in sulfate concentrations was observed at higher k_x values (Fig. 6b). These suggested that promoting sulfate reduction reactions through environmental manipulation could significantly accelerate the treatment process.

When the sulfate concentration was increased the rate of formation of insoluble metal sulfide products increased, which significantly accelerated the decrease in HM concentration (Fig. 6c and d). This is important for improving the treatment efficiency in wastewater treatment or HM pollution management.

Compared with previous SRB-based AMD treatment studies, this work demonstrated a much higher removal efficiency (100 %) for both Fe^{3+} and Zn^{2+} , with faster pH and Eh stabilization [24]. While many prior models have described SRB kinetics empirically, few have provided a full stability analysis with validated ODE systems [18,22]. Our approach combines experimental verification and mathematical modeling to deliver both accuracy and predictability, bridging the gap

between laboratory observation and reactor design optimization.

3.5. Discussion: model advantages, innovation, and practical implications

The developed ODE-based model addresses key limitations of experimental approaches by providing mechanistic insights and predictive capabilities that are difficult to obtain through observation alone [18]. Notably, the model enables long-term simulation of SRB dynamics and treatment outcomes, which is essential for assessing the sustainability of AMD remediation systems. It also facilitates operational optimization by identifying ideal ranges for sulfate, organic carbon, pH, and temperature to maximize SRB activity and reduce treatment costs. Furthermore, the model quantifies the interplay between SRB population dynamics, heavy metal (HM) sequestration, and environmental parameters, thereby offering a deeper understanding of the mechanisms underlying treatment performance.

Compared to previous studies that primarily reported empirical HM removal using SRB, this work presents two notable advancements. First, while past research has demonstrated the bioprecipitation of Fe^{3+} and Zn^{2+} via SRB metabolism [5,6], our system achieved complete (100 %) removal, which is explicitly linked to the in situ biosynthesis of metal sulfides—a process rarely quantified in prior literature [4]. Second, unlike earlier models that lacked predictive power, our study integrates a validated ODE system with stability and sensitivity analyses [22], providing a robust framework for system design and control.

3.5.1. Industrial scaling and system stability

Scaling up SRB-based systems for industrial application presents engineering challenges such as maintaining strict anaerobic conditions, ensuring uniform substrate distribution, and minimizing biomass washout under high flow rates [5,6]. A modified Up-flow Anaerobic Sludge Blanket (UASB) design may address these issues by enhancing biomass retention and mass transfer. The mathematical model developed herein can guide system optimization across variable hydraulic and loading conditions [18]. Future research should focus on pilot-scale validation and model-assisted process control to enable full-scale deployment.

Additionally, the observed stability of SRB activity across multiple cycles suggests strong operational resilience, reducing the need for reinoculation or frequent intervention. This is particularly valuable for decentralized or remote remediation sites. The model's predictive power also allows for real-time monitoring and automated control [18], further supporting its application in intelligent bioreactors.

3.5.2. Model limitations and environmental variability

The current model assumes constant environmental conditions (e.g., temperature, substrate levels, and microbial community structure). In reality, AMD systems are subject to seasonal fluctuations, competitive microbial interactions, and geochemical variability [1,3]. Although these factors were not explicitly simulated, the ODE framework can be extended to incorporate temperature-dependent kinetics and multi-population interactions [12]. Future studies should test model robustness under dynamic conditions using stochastic or agent-based approaches and validate against field data.

3.5.3. Integration with existing infrastructure

The proposed SRB system is compatible with conventional AMD treatment infrastructure [1,2]. It can function as a pre-treatment stage to reduce sulfate and metal loads before neutralization, or as a polishing step after chemical sedimentation [12]. Its modular and compact design allows for flexible retrofitting, and the model can support integration with sensor-based monitoring and feedback-controlled optimization.

3.5.4. Toward field validation

While this study was conducted under laboratory conditions, real-world AMD environments involve complex factors such as soil–water interactions, native microbial competition [1], and fluctuating geochemical parameters. To bridge this gap, pilot-scale trials at AMD-contaminated sites are essential [18]. These trials will not only validate SRB performance and metal sulfide formation [4] but also allow for refinement of model parameters and assessment of long-term feasibility under practical constraints.

4. Conclusion

This study integrates laboratory-scale AMD treatment with mathematical modeling to evaluate the performance and stability of SRB in removing HMs. Key findings were 100 % removal of Fe^{3+} and Zn^{2+} , accompanied by a beneficial increase in pH from 5.5 to 8.0, a ~50 % reduction in TOC, and a decline in redox potential from +86.8 mV to –200 mV. SEM imaging confirmed the in situ biosynthesis of metal sulfide, and microbial analysis showed a shift toward SRB-dominated communities.

Building on the experimental results a novel ODE model was developed and validated against experimental data, accurately simulating microbial growth, sulfate reduction, and metal sequestration. Equilibrium point analysis and sensitivity results highlight the potential for long-term system stability and parameter-guided optimization. Compared to previous studies, this work offers an enhanced, model-based understanding of SRB-mediated AMD treatment and bridges the gap between empirical observation and predictive control.

Collectively, these results provide a scalable framework for

intelligent bioreactor design and operation. Future work should focus on pilot-scale validation under field conditions and the integration of model-based automation to support decentralized and sustainable mine wastewater remediation.

CRediT authorship contribution statement

Jinyang Chen: Writing – original draft, Investigation, Formal analysis, Data curation. **Miqin Chen:** Writing – review & editing, Supervision, Project administration, Formal analysis. **Wensheng Yang:** Methodology, Formal analysis. **Li Gan:** Validation, Resources. **Gary Owens:** Writing – review & editing, Software. **Zuliang Chen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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